

Group	n/a
IRIS ID	[to be assigned]

Design: APOE p.R176C

Strategy document generated by ASSURED CRISPR Designer.

Edit class
SNP knock-in

Target
R176C

Guides
1

Primers
~470 bp

1. Gene Information

GENE
APOE

DESIGN CLASS
Point mutation

TARGET
APOE p.R176C | requested:
APOE R176C SNP knock-in

CELL LINE
n/a

DELIVERY
Synthetic guide / Cas RNP

TRANSCRIPT
NM_000041.4

PROTEIN / CDS
n/a

REFERENCE
apoe-r154s.gb

2. gRNA Sequences

Name	Sequence	Strand	GC	Notes
APOE_R176C_gRNA1	CGATGACCTGCAGAAGCGCC TGG	+ strand	65%	Cut-to-edit distance 1 bp preferred window <=10 bp exonic (exon 4): Guide spans 23 bp within exon 4.

3. Validation Primers

APOE_R176C_Fw18 nt

TGGAGGAACAACTGACCC

TM
59.5 C

GC
55.6%

CLAMP
3/3

APOE_R176C_Rev18 nt

CTCCTTCACCTCGTCCAG

TM
60.1 C

GC
61.1%

CLAMP
2/3

Expected amplicon: ~470 bp

Primer strategy: validated-centered

Alternative Validated Primer Pairs

Rank	Forward	Fw Tm	Fw GC	Fw Clamp	Reverse	Rev Tm	Rev GC	Rev Clamp	Amplicon
#2	TGGAGGAACAACTGACCC	59.5 C	55.6%	3/3	CACCTGCTCCTTCACCTC	60.7 C	61.1%	2/3	476 bp
#3	TGGAGGAACAACTGACCC	59.5 C	55.6%	3/3	CTGCTCCTTCACCTCGTC	60.7 C	61.1%	2/3	473 bp
#4	TGGAGGAACAACTGACCC	59.5 C	55.6%	3/3	CCTGCTCCTTCACCTCGT	62.0 C	61.1%	2/3	474 bp
#5	CTGGAGGAACAACTGACCC	61.0 C	57.9%	3/3	CACCTGCTCCTTCACCTC	60.7 C	61.1%	2/3	477 bp

DESIGN READINESS

Reference anchored
Design is anchored to a concrete reference sequence or validated lookup.

Guide expression compatibility

Pass

Review

APOE_R176C_gRNA1: GC 65% is outside the preferred 40-60% guide window.	
Requested edit validated R176C was mapped onto the coding sequence.	Pass
Guide geometry acceptable Selected 1 guide with best cut distance 10 bp from the edit.	Pass
Guide blocking strength APOE_R176C_gRNA1: strong (TGG->TCG removes the NGG PAM with a C/T substitution at PAM position 2 or 3.)	Pass
Final donor protein Every emitted donor translates to the intended R176C product without additional amino-acid changes.	Pass
Guide/donor pairing Each ssODN strongly disrupts every offered guide target.	Pass
Primer thermodynamics Thermodynamic primer checks pass (~470 bp; Tm delta 0.6 C).	Pass
Primer genome specificity Genome-wide primer specificity is not automatic; run the specificity check before ordering.	N/A

TARGET REGION MAP

Showing 2182-2701 on the uploaded reference sequence.

Exon / intron structure

Exon 4

Guides, edits, donors, and primers

Edit: R176C

Fw primer: TGGAGGAACAACCTGACCC

Rev primer: CTCCTTCACCTCGTCCAG

4. ssODN Donor Templates

WT and donor templates are listed together for review.

ssODN1 (matched to APOE_R176C_gRNA1) (- strand target)

Linked guide: APOE_R176C_gRNA1

Final donor protein assertion: Pass: final donor encodes C at residue 176 with no unintended coding changes.

Protection against all offered guides:

APOE_R176C_gRNA1: strong — TGG->TCG removes the NGG PAM with a C/T substitution at PAM position 2 or 3.

Silent mutation:

p.L177L: CTG -> CTC | PAM TGG->TCG

- strand donor

Order this strand

Recommended to order. This strand is reverse complement to APOE_R176C_gRNA1. Cut site lies between the 91 bp and 36 bp arms.

91 bp arm (91 nt)

36 bp arm (36 nt)

gRNA site

PAM

Desired edit

Silent mutation

WT

CCCTGTTCCACCAAGGGCCCCAGGCGCTCGCGGATGGCGCTGAGGCCGCGCTCGGCGCCCTCGGGGCCCCGGCCTGGTACACTGCCAGGCGCTTCTGCAGGTCATCGGCATCGCGGAGGAGCCGCT

Donor

CCCTGTTCCACCAAGGGCCCCAGGCGCTCGCGGATGGCGCTGAGGCCGCGCTCGGCGCCCTCGGGGCCCCGGCCTGGTACACTGCCAGGCGCTTCTGCAGGTCATCGGCATCGCGGAGGAGCCGCT

+ strand donor

Reference strand

Opposite donor strand for reference. Cut site lies between the 36 bp and 91 bp arms on this view.

36 bp arm (36 nt)

91 bp arm (91 nt)

gRNA site

PAM

Desired edit

Silent mutation

WT

AGCGGCTCCTCCGCGATGCCGATGACCTGCAGAAAGGCCCTGGCAGTGTACAGGCCGGGGCCCGAGGGCGCCGAGCGCGGCCTCAGCGCCATCCGCGAGCGCCTGGGGCCCCCTGGTGGAAACAGGG

Donor

AGCGGCTCCTCCGCGATGCCGATGACCTGCAGAAAGTGCCTGGCAGTGTACAGGCCGGGGCCCGAGGGCGCCGAGCGCGGCCTCAGCGCCATCCGCGAGCGCCTGGGGCCCCCTGGTGGAAACAGGG

CODING FRAME VIEW

WT codons																											
CGG	CTC	CTC	CGC	GAT	GCC	GAT	GAC	CTG	CAG	AAG	CGC	CTG	GCA	GTG	TAC	CAG	GCC	GGG	GCC	CGC	GAG	GGC	GCC	GAG	CGC		
GGC	CTC	AGC	GCC	ATC	CGC	GAG	CGC	CTG	GGG	CCC	CTG	GTG	GAA	CAG													
Donor codons																											
CGG	CTC	CTC	CGC	GAT	GCC	GAT	GAC	CTG	CAG	AAG	TGC	CTC	GCA	GTG	TAC	CAG	GCC	GGG	GCC	CGC	GAG	GGC	GCC	GAG	CGC		
GGC	CTC	AGC	GCC	ATC	CGC	GAG	CGC	CTG	GGG	CCC	CTG	GTG	GAA	CAG													
WT amino acids																											
R	L	L	R	D	A	D	D	L	Q	K	R	L	A	V	Y	Q	A	G	A	R	E	G	A	E	R		
G	L	S	A	I	R	E	R	L	G	P	L	V	E	Q													
Donor amino acids																											
R	L	L	R	D	A	D	D	L	Q	K	C	L	A	V	Y	Q	A	G	A	R	E	G	A	E	R		
G	L	S	A	I	R	E	R	L	G	P	L	V	E	Q													

Desired edit 1: genomic position 2442, C->T
APOE_R176C_gRNA1: p.L177L (CTG -> CTC) | PAM TGG->TCG

5. Review Checkpoints

- **Warning:** IRIS/internal project ID is missing. Assign it before release so the design can be joined to downstream clone-verification records.
- **Warning:** Requesting group is missing from the provenance metadata.
- **Check:** Record transcript assumptions, exon numbering assumptions, and delivery method before final sign-off.
- **Warning:** APOE_R176C_gRNA1: GC 65% is outside the preferred 40-60% guide window.
- **Check:** Confirm the desired amino-acid change against the intended transcript and verify that the donor does not create unwanted amino-acid substitutions.

6. Additional Info

Design: APOE p.R176C
Codon: CGC -> TGC
Guide/donor use: Each ssODN strongly disrupts every offered guide target.

gRNAs:
- APOE_R176C_gRNA1: CGATGACCTGCAGAAGCGCC TGG | + strand | GC 65%

Silent mutations:
- APOE_R176C_gRNA1: p.L177L (CTG -> CTC) | PAM TGG->TCG

Validation primers:
- APOE_R176C_Fw: TGGAGGAACAACTGACCC
- APOE_R176C_Rev: CTCCTTCACCTCGTCCAG
Expected amplicon: ~470 bp
Primer QC: high confidence | pair penalty 11.7 | Tm delta 0.6 C